1. Transport Layer & Data link Layer

Given:

File size = 150MB

Bandwidth = 50Mbps

Segment size = 1500 bytes

Header + Trailer = 40 bytes/frame

$$\text{Segments} = \frac{150{,}000{,}000}{1500}$$

Segments = 100,000

Total Overhead = 100,000 × 40

= 4,000,000 bytes

= 4MB

$$\text{Transmission Time} = \frac{1200}{50}$$

= 24 sec

Result:

Segments = 100,000

Total Overhead = 4MB

Transmission Time = 24 sec.

2. Sliding Window

Given:

Window size = 6

Total frames = 48

Frame size = 1024 bytes

$$\text{Transmission Windows} = \frac{\text{Total frames}}{\text{Window size}}$$

$$= \frac{48}{6}$$

$$= 8$$

Retransmissions = $8 \times 1 = 8$.

One frame lost in every window

Result:

Transmission Windows = 8

Retransmissions = 8

→ Flow control prevents the sender from overwhelming the receiver, reducing congestion and improving efficiency.

3. IP Fragmentation

Given:

Packet = 12,000 bytes

MTU = 1500 bytes

Header = 20 bytes

fragment = 1500 − 20 = 1480 bytes

$$\frac{12000}{1480} = 8.1$$

$9 \times 20 = 180$ bytes.

→ The Network Layer fragments large packets and reassembles them at the destination using identification numbers and fragment offsets.

4. Sliding Window.

Transmission Windows = $\frac{48}{6} = 8$

Retransmissions = $8 \times 1 = 8$

→ Regulates transmission speed and avoids receiver buffer overflow.

5. Session Layer

Given :

Video = 600MB

Checkpoint every 60MB

failure after 390MB

Checkpoints = 6

390 − 360 = 30MB → Retransmitted data.

Checkpoints before failure

60, 120, 180, 240, 300, 360.

→ Synchronization checkpoints allow communication to resume from the last checkpoint instead of retransmitting the entire file.

6. Presentation Layer.

Given:

Original = 900MB

Compression = 40%.

Encryption Overhead = 5%.

Compressed size = 900 × 0.60 = 540MB

Encrypted size = 540 × 1.05 = 567 MB

Reduction = 900 − 567 = 333MB

$$\text{Percentage} = \frac{333}{900} \times 100 = 37\%$$

The presentation Layer compresses data to reduce transmission size and encrypts it for confidentially.

7: FTP Application:

Given:

Data = 3GB

Throughput = 120Mbps

Request size = 3MB

3GB = 3000MB

Requests = 3000/3 = 1000

3GB = 24000 Mb

Transmission time = 24000/120 = 200sec

→ The Application layer provides services like FTP, file transfer, authentication and user interaction

8. End - End Delay

Given:

Application = 4ms , propagation = 18ms

Transport = 3ms , Transmission = 12ms

Network = 5ms

Data link = 2ms

Physical = 1ms

$$Total = 4+3+5+2+1+18+12 = 45ms$$

Total delay = 45ms

→ Each OSI layer performs specific processing before data is transmitted.

Protocol Overhead.

Given:

Useful data = 80MB

Frame size = 1600 bytes

Overhead = 80 bytes

$$Frames = 80,000,000 / 1600 = 50,000$$

$$\text{Total Overhead} = 50,000 \times 80$$

$$= 4,000,000 \text{ bytes}$$

$$= 4MB$$

Efficiency,

$$\text{Useful payload} = 1600 - 80 = 1520$$

$$\frac{1520}{1600} \times 100 = 95\%$$

10. ## Multiple OSI layers

Given:

Original file $= 240MB$

compression $= 30\%$

segment size $= 1400$ bytes

frame overhead $= 60$ bytes

Compressed size $= 240 \times 0.70 = 168MB$

$$\text{segments} = \frac{168,000,000}{1400} = 120,000$$

Protocol Overhead

$= 120,000 \times 60 = 7,200,000$ bytes

$= 7.2MB$

transmitted data $= 168 + 7.2 = 175.2MB$.